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CYBER DEFENSE RESEARCH INITIATIVE

An Intelligent Agent for Phishing Simulation and Behavioral Threat Analysis

4	3	7	12
BEHAVIORAL Profiles Modeled	ARCHITECTURE Tiers Integrated	ATTACK SCENARIOS Evaluated in Suite	STAGE AUTOMATED Execution Workflow

PROJECT TEAM

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ABSTRACT

Generative AI-driven spear-phishing, credential harvesting and MFA bypass outpace static, template-based security awareness training. **SemanticShield AI** is an agentic platform that uses LLMs to simulate how employees with four psychological profiles react to phishing lures, while a deterministic RiskEngine produces auditable threat scores (0–100). It adds live reasoning streams, automated SOC incident creation and real-time telemetry, with **zero real email delivery**.

`Agentic LLM Simulation` `Deterministic RiskEngine (0–100)` `Live Reasoning Telemetry` `Zero Real Email Delivery`

RESEARCH GAPS ADDRESSED

- GAP 01 Template-based simulations employees quickly recognize
- GAP 02 No psychological behavioral modeling of users
- GAP 03 Manual threat analysis causing SOC alert fatigue
- GAP 04 Training tools disconnected from SOC and browser defenses

AGENTIC AI PIPELINE: METHODOLOGY

Phishing Lure
Scenario, employee role, behavior profile, email subject and body

→

Simulation Engine
Central orchestrator
Google Gemini API

→

I Profile Selection
II SSE Reasoning Stream
III Decision Extraction
IV Composite Risk Score
V SOC Incident Creation
VI WebSocket Broadcast

→

Risk Verdict
VULNERABLE or SAFE, score, IoCs, stored in SQLite

DETERMINISTIC THREAT FORMULA `Composite Risk = 0.40 × Semantic + 0.25 × Behavior + 0.35 × URL`

PROFILES: Trusting · Security-Conscious · Overwhelmed · Executive

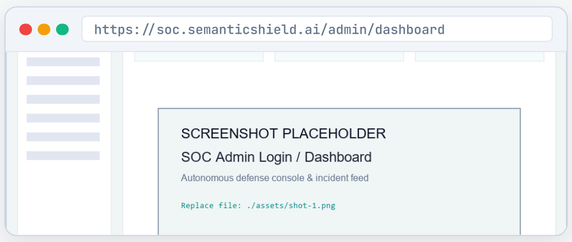
LAYERED SYSTEM ARCHITECTURE

Presentation LAYER 4	React 18 · Vite · TypeScript · Chrome MV3 extension
Gateway LAYER 3	FastAPI REST · Asyncio · Supervisor orchestrator
AI Integration COGNITIVE CORE	Gemini API · SSE reasoning streams
Analytics LAYER 1	RiskEngine · Explainability · SQLite · WebSocket

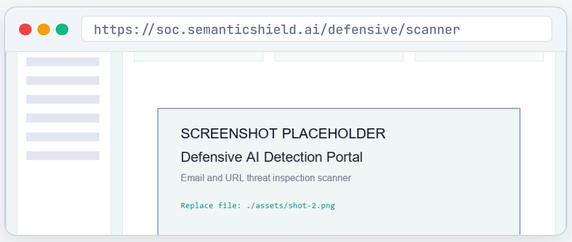
END-TO-END WORKFLOW

1. Receive Request START
2. Validate Fields
3. Select Profile
4. Gemini Stream
5. Extract Decision
6. Risk Scoring
7. Create Incident
8. Log Activity
9. Broadcast Event END

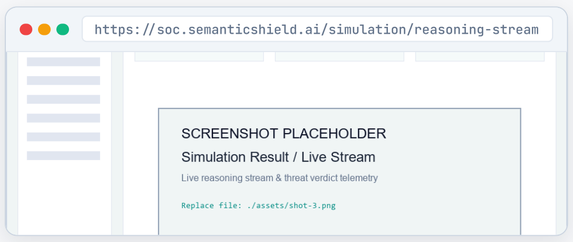
IMPLEMENTATION: WORKING PROTOTYPE



SOC portal: administrator login to the autonomous defense dashboard



Defensive AI Detection Portal: email and URL scanner



Simulation result: live LLM reasoning stream and threat verdict

28 RUNS EVALUATED

7 scenarios × 4 profiles. Credential harvest, wire scam, quishing, delivery, MFA bypass and a legitimate stand-up.

84/100 DANGEROUS	18/100 SAFE

AVERAGE LATENCY 1.42 s

KEY FINDINGS

- Trusting and Overwhelmed profiles fell for urgency cues
- Executive profile resisted low-level delivery and MFA lures
- Identical scoring for offensive and defensive scans
- Zero real-world email delivery risk

TECHNOLOGY STACK

- React 18
- TypeScript
- FastAPI
- Python 3.13
- Gemini API
- SQLite
- SQLAlchemy
- WebSockets
- SSE
- Chrome MV3

CONCLUSION

Separating non-deterministic LLM behavioral reasoning from mathematical risk scoring delivers realistic, educational and auditable phishing awareness, with real-time SOC integration and no real email delivery.

LIMITATIONS & FUTURE WORK

LIMIT:
The system depends on the Gemini API and simulated employee behavior, which may not fully reflect real-world human responses.

NEXT:
Integrate multiple AI models and deploy the system on cloud platforms for improved scalability and threat detection.